

Fig. 14: Summary of Model Performance Metrics

Model	Source	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Expression	Camera	93.92	94.05	93.69	93.86
Blink	Camera	83.08	84.11	83.74	83.85
Posture	Camera	86.12	86.27	86.33	86.19
Physio	MAX30102	87.92	87.99	88.55	88.18
PHQ-9	Questionnaire	92.40	92.41	92.44	92.42
Combined	All Sensors	98.28	98.33	98.20	98.27
Ensemble	All (Stacking)	98.28	98.33	98.20	98.27